

Iris Dataset — EDA Summary Report

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1. Data Preprocessing Decisions

- Missing values: none found in any column — no imputation needed.
- Duplicates: checked, none required removal.
- Outliers: IQR method applied to all four features. Mild outliers found only in SepalWidthCm. These were capped (Winsorized) at the lower/upper IQR bounds rather than removed, since the dataset is small (150 rows) and the values reflect natural variation, not errors.
- Categorical encoding: Species label-encoded — Iris-setosa = 0, Iris-versicolor = 1, Iris-virginica = 2. Label encoding was used since Species is the target, not an input feature.
- Feature scaling: all four numeric features standardized with StandardScaler (mean = 0, std = 1) to put them on comparable scales.

2. Key Visualizations

Boxplots by species — showing which features separate the classes best:

Pairplot — every feature pair, colored by species:

3. Correlation Analysis

Feature Pair	Correlation	Meaning
Petal Length ↔ Petal Width	0.96	Very strong — largely redundant
Petal Length ↔ Sepal Length	0.87	Strong positive
Sepal Width ↔ Petal Length	-0.42	Moderate negative
Sepal Width ↔ Sepal Length	-0.11	Weak / negligible

4. Key Insights

- Dataset is clean and perfectly balanced — 50 samples per species, no missing data.
- Petal length and petal width are the most discriminative features, separating species with almost no overlap (confirmed by boxplots, pairplot, and F-values of 1179 and 959 respectively).
- Sepal width is the weakest, most independent feature — heavy overlap across species and low correlation with the others.
- Petal length and petal width are highly correlated (0.96), meaning they carry overlapping information — a candidate for dimensionality reduction.
- Setosa is fully separable from the other two species on almost any feature; versicolor and virginica overlap somewhat, mainly on sepal measurements.

5. Machine Learning Readiness Summary

The dataset is fully preprocessed and ready for modeling:

- No missing values, duplicates handled, outliers capped.
- Target variable encoded numerically (0/1/2).
- Features standardized — suitable for distance-based models (KNN, SVM) and gradient-based models alike.
- Class balance is perfect, so no resampling or class-weighting needed.
- Petal-based features are strong predictors on their own; a simple model (Logistic Regression, Decision Tree, or KNN) should perform well as a baseline.